Tutorial Questions 02 – Series Dr Killian O'Brien
6G4Z9019 Mathematical Methods 23/10/26

When possible you should work on the questions in advance and be prepared for discussions on them with your colleagues and tutor. Some questions are straightforward calculation while others require more discussion and thought. The computer icon 🖳 indicates where computer calculation or programming may be of use (although it is potentially useful everywhere I guess!). Further questions can be found in the lecture notes.

Series notation and defining formulas

1. Write down some initial terms of the following series

$$\text{(i)} \sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1}, \quad \text{(ii)} \sum_{n=1}^{\infty} \frac{n+1}{n^3+2}, \quad \text{(iii)} \sum_{n=1}^{\infty} \frac{3}{n!}, \quad \text{(iv)} \sum_{n=1}^{\infty} \left(\sum_{i=1}^{n} i^{-2} \right).$$

Series convergence definition

2. Make sure you can accurately quote the definition of convergence for a series.
3. Can you confidently give the proof that the harmonic series $\sum_{j=1}^{\infty} 1/j$ diverges?
4. Can you prove the General Term Test using the definition of convergence?

Geometric series test

5. Ensure that you can derive the partial sum formula for a general geometric series of the form

$$\sum_{n=1}^{\infty} ar^{n-1}.$$

   Can you then prove the convergence/divergence of this series for all possible combinations of values of the initial term a and the common ratio r?
6. Use the analysis of the geometric series test to determine the convergence status and sum if possible, of the series

$$\text{(i)} \sum_{n=1}^{\infty} \frac{2}{3} \left(\frac{1}{3} \right)^{n+1}, \quad \text{(ii)} \sum_{n=1}^{\infty} 3 \left(\frac{1}{2} \right)^{n}, \quad \text{(iii)} \sum_{n=1}^{\infty} \frac{1}{5} \left(\frac{6}{5} \right)^{n+2}.$$

7. Consider the number a with decimal expansion

$$a = 0.1234123412341234\ldots,$$

   continuing in the suggested way. Use the geometric series summation formula to express a as a rational number, i.e. a ratio of integers.

Application of the ratio test

8. Determine the convergence or divergence of the series

$$\sum_{m=0}^{\infty} \frac{2^m}{m!}.$$

9. Determine the convergence or divergence of the series

$$\sum_{m=0}^{\infty} \frac{2^m}{(m+1)^2}.$$

10. Does the ratio test give a conclusion about the series

$$\sum_{n=1}^{\infty} \frac{n}{n+1}?$$

    If not, determine the convergence/divergence of this series by more direct means.

Application of comparison tests

11. Use a suitable comparison test to determine the convergence status of the series

$$\sum_{k=1}^{\infty} \frac{k^2+2}{k(k^2+5)}.$$

12. Use a suitable comparison test to determine the convergence status of the series

$$\sum_{k=1}^{\infty} \frac{k+3}{k(k^2+2)}.$$

Application of partial fraction expansions

13. Consider the series $\sum_{n=1}^{\infty} a_n$ where the term is defined by

$$a_n = \frac{9\,n^2+33\,n+12}{(n+4)(n+3)(n+1)n}.$$

Obtain the partial fraction expansion of a_n and use this to obtain the sum of the series.

14. Consider the series $\sum_{n=1}^{\infty} b_n$ where the term is defined by

$$b_n = \frac{8}{n^2+6n+5}.$$

Obtain the partial fraction expansion of b_n and use this to obtain the sum of the series.

15. Consider the series $\sum_{n=2}^{\infty} c_n$ where the term is defined by

$$c_n = \frac{9}{n^2+n-2}.$$

Obtain the partial fraction expansion of c_n and use this to obtain the sum of the series. (n.b. take note of the starting value of the index n)

16. Consider the series $\sum_{n=1}^{\infty} d_n$ where the term is defined by

$$d_n = \frac{11-n}{n^2+6n+5}.$$

Obtain the partial fraction expansion of d_n and use this to decide whether the series is convergent or divergent.

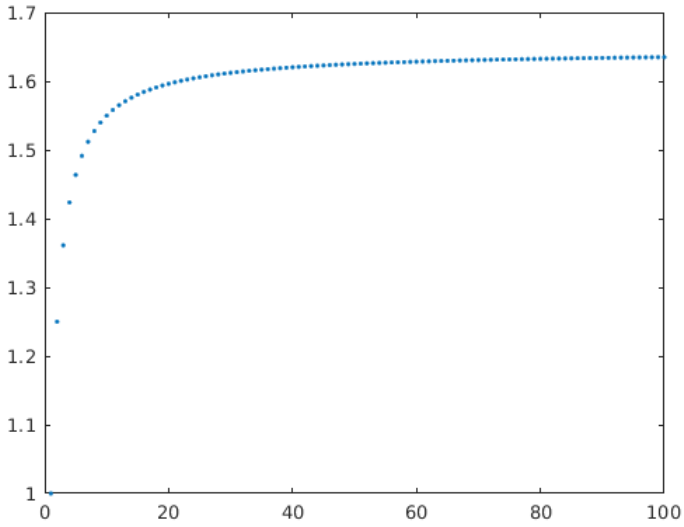
Computer work



17. You should ensure that you are able to use Matlab to build an array containing a specified number of initial terms of a series.

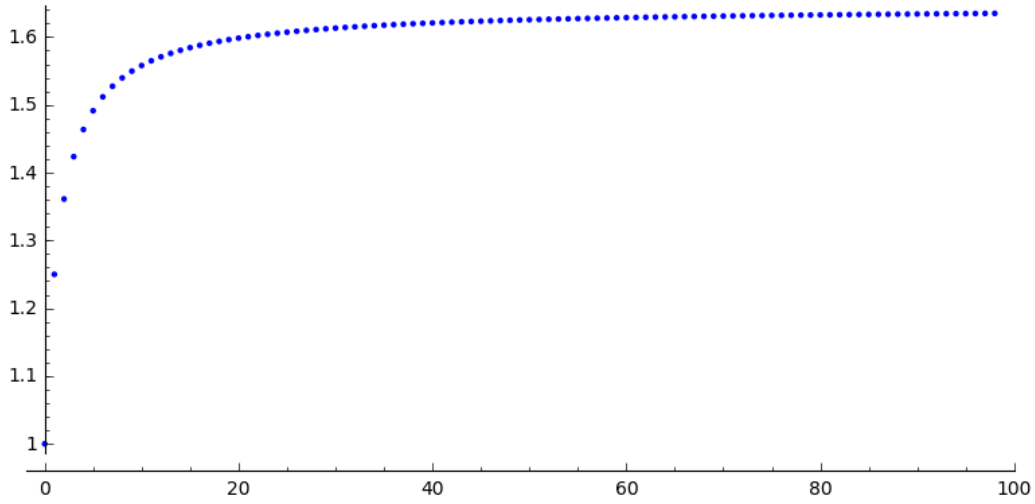
18. You should then be able to build an array containing the partial sums of that series and plot them as visual aid to determining the long term behaviour of the series.

```
x = @(n) 1/n^2 % creates a function handle x for the series term
N = [1:100] % the index values 1, 2, 3, ..., 100
Xseries = arrayfun(x,N) % a vector of values 1/n^2
Xpartials = cumsum(Xseries) % the cumsum function creates a vector of
    cumulative (partial) sums
plot(Xpartials,'.') % plots the partial sums
```



19. Investigate the capabilities of Matlab's `symsum` command from its symbolic toolbox. See the information at `http://uk.mathworks.com/help/symbolic/symsum.html`.
20. Apply these techniques to the series mentioned in this problem sheet, the questions in the lecture notes, etc
21. Investigate the capabilities of other computer systems, such as SageMath in this regard. Register for a free account at `https://cocalc.com` and check out Gregory Bard's book *Sage for Undergraduates* available from `http://www.gregorybard.com/Sage.html`.

```
1 Xseries = [1/n^2 for n in range(1,101)]
2 Xpartials = [sum(Xseries[:k]) for k in range(1,100)]
3 list_plot(Xpartials)
```



Further problems

In these problems there is not so much guidance given about which particular convergence test to use. Sometimes it is a process of trial and error but with experience and understanding of the different tests one can often identify the appropriate test from the form of the given series.

22. Investigate and prove the convergence/divergence of the following series.

$$\text{(i)} \sum_{n=1}^{\infty} \frac{n^3}{n^5+2} \qquad \text{(ii)} \sum_{n=1}^{\infty} \frac{3^n}{5^n+5} \qquad \text{(iii)} \sum_{n=1}^{\infty} \frac{n}{3^n}.$$

23. The series $\sum_{n=1}^{\infty} n^a/(2+n^3)$ depends on the value of the parameter a. For which values of a will the series converge? For which values of a will it diverge?

24. Investigate and prove the convergence or divergence of the series

$$\sum_{n=1}^{\infty} \frac{n!(n+1)!}{(3n)!}.$$

25. Investigate and prove the convergence or divergence of the series

$$\sum_{n=1}^{\infty} (-1)^n \cos\left(\frac{1}{n}\right).$$

26. Consider the power series

$$\sum_{n=0}^{\infty} \frac{n^3}{n^4+1} x^{3n}.$$

What is the radius of convergence of this power series?

27. Can you find the precise value of the series

$$\sum_{n=1}^{\infty} \frac{1+2^n}{3^{n-1}}?$$

Hint: Which series convergence test(s) allow you to find the actual values of infinite sums?

28. Does the series $\sum_{n=1}^{\infty} \frac{n+5}{n\sqrt{n+3}}$ converge?

29. Does the series $\sum_{n=1}^{\infty} \frac{3+\cos(n)}{e^n}$ converge?

30. Does the series $\sum_{n=0}^{\infty} (-1)^n (n^2+1)^{-1/2}$ converge? Does it converge absolutely? *Hint: The presence of the $(-1)^n$ terms means we should apply the _____ test*

31. What is the value of the sum

$$\sum_{n=1}^{\infty} \frac{2^n+5^n}{7^n}?$$

32. Will the series

$$\sum_{n=1}^{\infty} (-1)^n \frac{n}{1+n^2}$$

converge? Will it converge absolutely?